

SeaIce Classification Deep Fusion Workflow

Project Overview and Implementation Plan

This document outlines a comprehensive workflow for seaice classification using multimodal data (SAR imagery and tabular ICESat2 features). The updated approach leverages a UNet encoder and an LSTM network for deep fusion.

1. Data Overview & Objectives

The sea-ice classification problem is formulated as a per-sample prediction task. Each sample consists of a raw SAR image patch and a set of tabular features.

- **SAR image patches:** High-resolution Sentinel-1 Ground Range Detected (GRD) products are preprocessed (speckle filtering, radiometric calibration and geometric correction) and then resampled to 10 m resolution. Local patches of size 128×128 pixels are extracted around each ICESat-2 photon point (632541506862653†L807-L820). These patches contain texture and spatial information relevant for sea-ice identification.
- **CSV-based altimetry features:** Each row of the ICESat-2 ATL03 product provides photon height statistics (mean, median, standard deviation), reflectance and beam information, together with geographic coordinates and a manually assigned sea-ice label. These features capture physical properties of the ice and water surfaces. Each CSV entry is matched to a corresponding SAR patch based on its pixel coordinates (pix_x, pix_y). The goal is to classify each pair into one of the target classes (e.g., open water, young ice, thick ice). This fusion of heterogeneous data sources allows complementary information to improve classification accuracy (632541506862653†L701-L716).

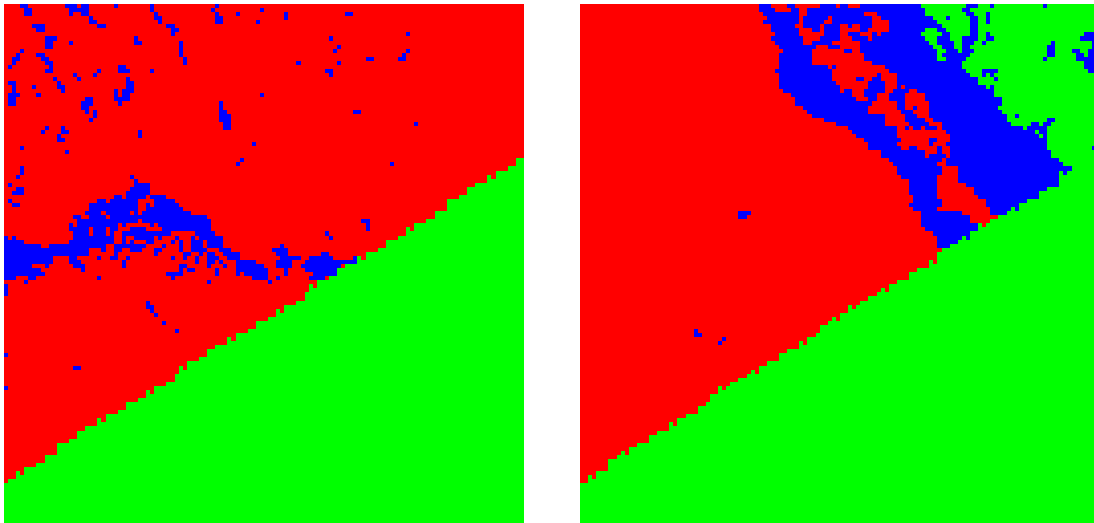


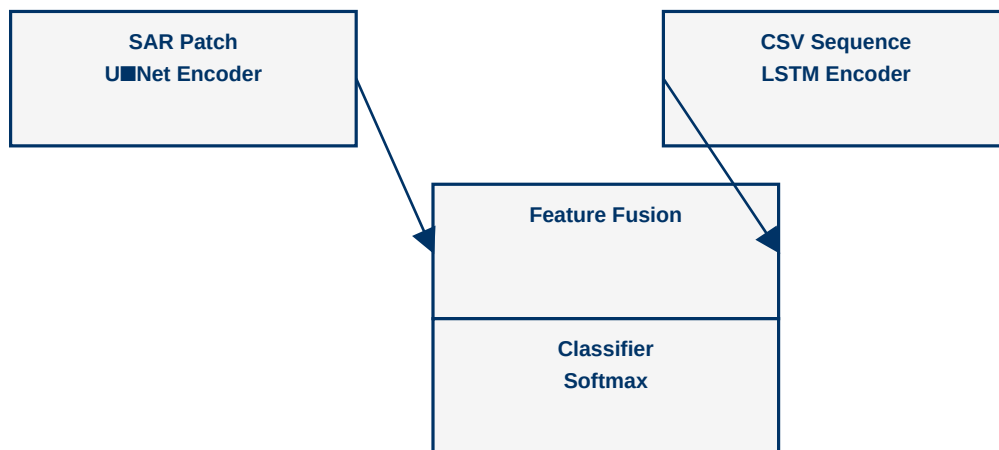
Figure 1: Examples of SAR patches extracted around labelled ICESat-2 photon points

2. Baseline Models

Establishing single-modality baselines clarifies the contribution of each data source. **CSV-only baseline:** A feedforward multilayer perceptron (MLP) takes the standardized numerical features (height statistics, reflectance, beam ID, etc.) and predicts the ice class. Performance metrics (overall accuracy, F1 score and IoU) gauge the discriminative power of tabular features alone. **SAR-only baseline:** A convolutional neural network (CNN) processes the raw 128×128 SAR patch. Layers of convolution and pooling, followed by global average pooling and dense layers, produce a class prediction. This baseline measures the value of texture and spatial patterns. Comparing these baselines to the deep fusion model quantifies the gain from integrating both modalities.

3. Deep Fusion with U-Net & LSTM

In the deep fusion architecture, two specialized encoders process the SAR images and ICESat-2 features before merging their outputs. **U-Net encoder for SAR patches:** A U-Net model captures multi-scale spatial features from the 128×128 SAR patch. Only the encoder portion is used for classification: convolutional blocks and downsampling layers extract hierarchical texture information, while skip connections preserve fine-grained details. The encoded features are flattened to a fixed-length vector. **LSTM encoder for tabular sequences:** Instead of treating each CSV row independently, a sequence of neighbouring points along the satellite track is assembled (e.g., ±5 samples). An LSTM (long short-term memory) processes this sequence and learns temporal and spatial context. The final hidden state provides a vector summarizing the neighbourhood dynamics. **Feature fusion & classification:** The U-Net and LSTM vectors are concatenated and fed into fully connected layers with dropout regularization. A softmax classifier outputs the probability of each sea-ice class. Using class weights or balanced sampling helps mitigate label imbalance.



[1] SAR & optical feature extraction and fusion framework

[2] Two-branch network with multi-scale extraction and fusion

4. Implementation Workflow

The following steps outline a practical implementation of the U-Net+LSTM deep fusion model: **Step 1 – Data preprocessing:** Preprocess SAR images (speckle filtering, radiometric calibration, geometric correction) and resample to 10 m resolution. Normalize pixel values (e.g., log transform and z-score). Clean CSV data (remove invalid rows, handle missing values) and standardize numerical features. **Step 2 – Patch extraction & sequence assembly:** For each CSV row, extract a 128×128 SAR patch centered at (pix_x, pix_y). To construct LSTM sequences, gather neighbouring points along the satellite track (e.g., ±5 rows). Each sequence is an ordered list of feature vectors and labels. **Step 3 – Train baseline models:** Train the CSV-only MLP and SAR-only CNN to establish benchmarks. Evaluate using class-weighted loss to address label imbalance. **Step 4 – Train U-Net encoder:** Train a U-Net segmentation network on SAR patches and their labels to learn multi-scale representations. Discard the decoder during fusion: use the final encoder feature map, apply global average pooling and a dense layer to obtain a feature vector. **Step 5 – Train LSTM encoder:** Feed sequences of tabular features into an LSTM. Use the final hidden state as the sequence representation. Apply dropout to prevent overfitting. **Step 6 – Feature fusion & classification:** Concatenate U-Net and LSTM vectors. Pass through fully connected layers with dropout and softmax activation. Use class weights to handle imbalance. Evaluate performance using accuracy, precision, recall, F1 score and IoU. **Step 7 – Hyperparameter tuning & evaluation:** Tune patch size, sequence length, learning rate and network depth. Compare performance against baselines. Perform cross-validation or hold-out testing on different dates/tiles to ensure generalization.

5. Enhancements & Conclusion

Potential enhancements: Beyond the U-Net+LSTM baseline, explore attention-based fusion where the network dynamically weighs modalities. Test Transformers for tabular sequences or 3D-CNNs for larger patches. Incorporate additional inputs (e.g., optical Sentinel-2 images) to capture spectral information, similar to heterogeneous data fusion approaches [63, 254, 150, 686, 265, 37, 701-716]. **Conclusion:** Deep fusion of SAR and ICESat-2 data leverages complementary spatial and physical information. A U-Net encoder extracts rich multi-scale texture features from SAR patches, while an LSTM summarises context from neighbouring photon measurements. Through careful preprocessing, balanced training and comprehensive evaluation, this workflow aims to achieve robust sea-ice classification and paves the way for future multi-sensor extensions.

[1] Feature extraction and fusion components